White dwarf binaries: formation, evolution and stability

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ABSTRACT

White Dwarf (WD) binaries are important in stellar evolution; acting as Type Ia Supernova progenitors, and gravitational wave sources. Despite their importance however, the formation pathway of WD binaries (in particular long period WD-Main sequence and Double WD binaries) are poorly constrained. This review covers the current understanding of the Common Envelope (CE) and Stable Roche Lobe Overflow (SRLOF) mass transfer phases, looking in-depth at the debate between the α_{CE} and γ formalisms for CE. Recent developments like the development of eccentric mass transfer models and hybrid CE and SRLOF phases have been proposed to solve issues raised by the individual phases. This review highlights the need for both phases in order to develop the diversity seen in the current WD binary population.

1 INTRODUCTION

White dwarf binaries are binary systems that typically contain an accretor White Dwarf star (WD) and a donor main sequence star (MS). In some cases, white dwarf binaries can contain two white dwarf stars, known as a double white dwarf (DWD) binary. WD binary systems have been identified to have a range of orbital periods, spanning from a few minutes (Roelofs et al. 2010) to days (Iben & Livio 1993).

WD binaries are very useful in understanding the evolution of stars and the universe. Both single WD and double WD binaries can act as type Ia supernova (SN Ia) progenitors, should enough mass have transferred to a WD, it passing over the Chandrasekhar limit, and leading to the eventual collapse of these system into a SN Ia. SN Ia have previously been thought to act as standard candles and were used to determine the universe's rate of acceleration (Riess et al. 2004). WD binaries generate gravitational waves (GW), allowing them to be used as GW sources for detection, as the GW emitted are easier to detect than EM wave sources. GW are also emitted by DWD, meaning that newer GW detectors like LISA could detect the GW emitted by these systems, allowing for easier identification and potentially understanding their mechanisms (Amaro-Seoane et al. 2017; Colpi et al. 2024).

In the current understanding of WD binaries, there are a few problems not yet understood. It is currently unknown how double WD binaries are formed, due to the lack of observational data we have on these systems, be it from the faintness of these systems, the difficulty identifying WD binaries or the lack of understanding of the mass transfer history of these binary.

This review looks into the concepts of mass transfer in these types of WD binaries, common envelope evolution, Roche Lobe overflow stability, and potential solutions to the issues in these processes.

2 MASS TRANSFER IN BINARIES

Mass transfer is the process of mass leaving the donor star in the binary and the other accreting that mass. This transfer arises from the filling and expelling of the Roche Lobe's (RL) of the binary objects. Figure 1 shows a diagram of an example binary's Roche Lobes. A Roche lobe is the area surrounding a star that contains gravitationally bound orbiting material.

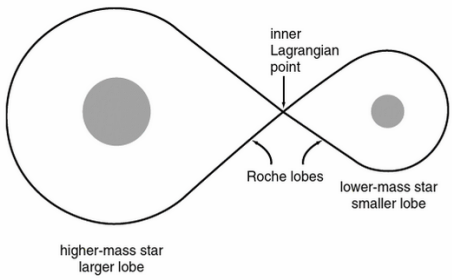

Figure 1. Example diagram of a binary's Roche Lobes. Taken from Wheeler (2007).

For most WD and MS star binaries, the binary objects begin as two MS stars where one evolves into a WD. When the MS star evolves to the RGB phase, it expands and can fill it's RL. If the MS star radius exceeds the RL radius, that mass is no longer gravitationally bound and can leave the star and is ejected into the binary. If this ejected mass passes the inner Lagrangian point, L_1, it enters the RL of the companion star. The mass transfer process and how it affects the RL's of the binary is discussed in later sections.

2.1 WD binary population

The population of known WD binary systems varies depending on the type of companion secondary star. Due to the inability to map the exact, complete WD+MS population of the universe, this review will refer to samples when discussing the population number of differing WD+MS binaries with differing secondary star stellar classes and secondary stars.

Rebassa-Mansergas et al. (2016) identified 938 WD+M-type binaries from the Sloan Digital Sky Survey (SDSS) data releases 9 to 12, resulting in a sample size of 3294 close WD+M-type binaries.

For WD+FGK binaries, two papers are highlighted in particular, Shahaf et al. (2024) and Garbutt et al. (2024) (notably these are the datasets used in the project surrounding this review). Garbutt

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et al. (2024) identified 206 WD+FGK binaries from ultraviolet excess catalogues, and Shahaf et al. (2024) identified 3145 WD+FGK binaries from "no-colour-excess" astrometric binaries from the Gaia DR3 catalogue (note this is the corrected number given in Shahaf et al. (2025)).

Both WD+M-type and WD+FGK binaries can be difficult to identify due to the higher luminosity of the MS star outshining the WD star, which can make the binary appear as a single luminous star if it were placed on a Colour-Magnitude diagram.

Munday et al. (2025) displayed a sample of 15 DWD binaries, with other papers like Nelemans et al. (2001) predicting a galactic population of approximately 2.5×10^8 DWD binaries.

The following sections will discuss common envelope evolution and stable Roche lobe overflow.

3 COMMON ENVELOPE EVOLUTION

3.1 Formation of a common envelope

In a giant star + MS binary, the more massive donor star fills its RL and begins mass transfer when the RL overfills and any material outside the RL is ejected. This causes the orbit of the binary to tighten, the RL to shrink and the stars radius to overflow the RL again. This constant filling and shrinking of the RL causes the mass-loss timescale to become so short that the MS donor cannot stay within its RL, a runaway process of dynamically unstable mass transfer. If the accretor cannot take all the mass being accreted onto it, it's RL fills quickly, therefore the excess mass forms a common envelope (CE) around the binary, engulfing both objects (Paczynski et al. 1976).

With the binary in a CE phase, the binary objects initially continue orbital rotation inside the envelope at a different speed to the envelope's rotation. This imbalance in speed causes the stars to experience a headwind (drag forces) inside the envelope gas, causing the objects lose angular momentum and energy. If sufficient energy is lost from the binary into the envelope, the envelope can be unbound. This unbinding causes a shorter period binary to emerge.

3.2 Energy (α_{CE}) formalism

The energy formalism, α_{CE} (Webbink 1984), was created to predict the outcome of common envelope evolution which is related to the ratio of the binding energy of the common envelope, E_{bind}, and the change in orbital energy, ΔE_{orb}. Belloni & Schreiber (2023) relates these values in the equation below. Equation 1 is generally parametrised by $0 \leq \alpha_{CE} \leq 1$, or the efficiency of the orbital energy used to expel the CE.

$$E_{bind} = \alpha_{CE} \Delta E_{orb} = -\alpha_{CE} \left(\frac{G M_{c,d} M_a}{2 a_f} - \frac{G M_d M_a}{2 a_i} \right) \tag{1}$$

where $M_{d,c}$ is the donor stars core mass, M_a is the accretor mass, a_i is the semimajor axis at the initial point of CE evolution and a_f is the semimajor axis at the expulsion of the envelope.

A small value of α_{CE} denotes a low efficiency in the orbital-to-binding energy ratio. Zorotovic et al. (2010) constrains α_{CE} and finds solutions for the sample to be $0.2 \leq \alpha_{CE} \leq 0.3$ for CE evolution of WD+M-type binaries.

3.2.1 Angular Momentum (γ) formalism

Nelemans et al. (2000) and Nelemans & Tout (2005) proposed an alternative formalism for constraining DWD binary CE evolution, the angular momentum, γ, formalism, as seen below in equation 2 (Zorotovic et al. 2010).

$$\frac{\Delta J_{lost}}{J_i} = \frac{J_i - J_f}{J_i} = \gamma \frac{m_{1,e}}{m_1 + m_2} \tag{2}$$

Where J_i, J_f and J_{lost} are the angular momentum of the initial and final binaries, and the angular momentum lost respectively, and $m_{1,e}$ is the mass of the expelled envelope.

Nelemans & Tout (2005) identified a value for γ between 1.5−1.75, which was believed to be able to construct nearly any observed binary, allowing for a high value of α_{CE}. However, this value for γ acted more like a property of equation 2, and did not hold up when the values were inverted in works by Webbink (2008) and Woods et al. (2010). Zorotovic et al. (2010) confirmed the findings of Webbink (2008), and agreed that in order to accurately predict the outcome of CEE, α_{CE} should be constrained to $0.2 - 0.3$, not γ.

3.3 Expected outcomes of CEE

In a general case, CEE forms short period binaries (from a few minutes to a few days) with a range of WD masses, which is the expected outcome for a successful post-CE binary. Successful post-CE binaries can also be cataclysmic variables (Paczynski et al. 1976), X-ray binaries (Podsiadlowski et al. 2002) and potentially type Ia supernova (Liu et al. 2023).

If the CE is not expelled, then a binary merger is expected, where the inward spiral of the objects continues to the centre of mass and merge. The outcome of this in-spiral often results in a singular rapidly rotating star or a Luminous Red Novae, where the merging of the binary objects release energy sufficient enough to be an observable luminosity change, such as V1309 Scorpii (Tylenda et al. 2011).

3.4 Issue with CEE

The issue with CEE is that if a successful ejection of the CE (with a low α_{CE}) allows for short, compact binaries to develop, and an unsuccessful ejection results in a merger of the binary objects, then how are wide, long period WD+FGK and DWD binaries formed, since a close binary is too close to survive a 2nd CE phase?

Belloni et al. (2024) analysed six long period WD+MS systems to see if the same period could be reconstructed without introducing external energy sources (apart from orbital, thermal and gravitational) into the CEE energy. The results produced the same orbital period result as the systems by allowing the WD progenitors to reach the TP-AGB star phase before filling the RL at the initial point of CEE. This works as the envelope mass and binding energy are reduced while the separation of the binary remains large, resulting in the envelope ejection requiring only small shrinkage and maintaining the long period/separation that the system was initally reported to have.

4 STABLE ROCHE LOBE OVERFLOW

The Roche lobe overflow (RLOF) process follows a similar structure to that outlined in section 2. If a donor MS star's surface surpasses its RL, the surrounding surface material can fall past the L_1 Lagrangian point between the binary object's Roche lobes and be accreted by the WD, transferring some of the donor star's angular momentum with it.

For SRLOF systems, as the donor MS star loses mass it reduces in size, faster than the RL shrinks in response and if the donor's

RL continues to surround the star, the RLOF remains stable (Tauris & van den Heuvel 2006). SRLOF occurs in binary systems with a radiative envelope MS donor star, where;

$$q = \frac{M_{donor}}{M_{accretor)}} \lesssim \frac{3}{5} \approx 0.634 \quad (3)$$

Value for equation 3 taken from Hjellming & Webbink (1987).

If the MS star has a convective envelope, the RLOF process is unstable and the outcome is more likely to follow the path of CEE.

Mass transfer can range between conservative and non-conservative. In the former, any and all mass lost by the donor star is accreted by the accretor which maintains the conservation of mass and angular momentum of the binary Parkosidis et al. (2026). No change in mass or angular momentum implies that any change in orbital separation is driven by the mass ratio, q.

In the latter, mass is lost from the system, meaning that mass is not conserved and in-turn, neither is angular momentum (Parkosidis et al. 2026) therefore, change in the orbital separation of the binary can come from mass or angular momentum loss, and lead to a tightening or potential merger of the binary objects.

4.1 Mass transfer cases

The type of binary dictates the type of mass transfer undergone, also known as Case A,B and C mass transfer, or the Kopal system (Kopal 1955).

Case A mass transfer involves a Hydrogen burning MS star, typically occurs in short period systems and is characterised by slow mass transfer lasting several 10^6 years.

Case B involves a MS star that has exhausted core H-burning, but before core Helium burning has completed, is most common in massive stars, and affects systems with an orbital period range of 5-1000 days.

Case C occurs once MS star has exhausted its core Helium supply, typically affecting the widest binaries with orbital period ranges of 1500-3000 days, and results in a "common envelope" phase.

4.2 Expected outcomes of SRLOF

Non-conservative mass transfer can produce long period WD binaries, due to the loss of angular momentum along with the mass lost from the binary system. Roy & Kalita (2026) completed simulations on DWD binaries, and under SRLOF model conditions, found an orbital separation range of $100 - 500 R_\odot$.

RLOF can also lead to a range of periods in binaries. Willems & Kolb (2004) completed evolutionary models that showed systems undergoing a SRLOF channel resulted in orbital periods ranging from 10s to approximately < 600 days. Mou et al. (2026) backs up this point of ranging orbital periods, by modelling the WD mass to orbital period ratio of post-SRLOF systems, showing that $M_{WD} \propto P_{orb}$.

Low mass WD stars can be formed from SRLOF in DWD systems. Sun & Arras (2018) shows that extremely low-mass WD stars ($0.2 \leq 0.4 M_\odot$) can be formed via SRLOF channels.

4.3 Issues with SRLOF

SRLOF can not make very short period systems, i.e. those on hour long timescales. This is due to factors like SRLOF needing to keep the mass ratio $q \leq 1$, and that a binary must lose very high orders of magnitude of angular momentum to tighten the binary to produce hour long period binaries - which is not possible to be lost in conservative SRLOF.

SRLOF also struggles to make high mass ($> 0.5 M_\odot$) WD stars, as Case C mass transfer is needed, which requires a MS star with a convective envelope. SRLOF (Case A and B mass transfer) also begins when the donor star is filling its RL on the MS, meaning that the mass transfer process strips the donors envelope before it has fused the necessary heavy elements needed for the WD to accrete them into its core to form a larger mass.

5 POTENTIAL SOLUTIONS

Solutions to the issues faced by CEE (inability to make long period binaries) and SRLOF (inability to make short period systems and low mass WD stars) have been proposed, like the previously mentioned solution to CEE by Belloni et al. (2024); where WD progenitors in WD+MS systems were modelled as TP-AGB stars at the initial point of CEE. Other solutions have also been proposed, which will be discussed in the following subsections; a combination of both SRLOF and CEE, and eccentric mass transfer.

5.1 A combination of non-conservative SRLOF and CEE

Halabi et al. (2018) proposes a combination of a CE phase, then RLOF phase, which allows for the development of a high mass WD ($\sim 0.8 M_\odot$) from a Hydrogen and Helium stripped MS donor star, and produces binaries with orbital period ranges of 10s of days to ~ 100 days.

Alternatively, Renzo et al. (2023) proposes a combination of RLOF, followed by a CE phase. This combination allows for the binding energy of the CE to be altered, making the expulsion of the envelope easier, and results in a wider separation binary, therefore allowing for a longer period binary than would be expected from a single CE phase.

DWD binaries can also be formed from a combination of RLOF and a CE phase Woods et al. (2012).

5.2 Eccentric mass transfer

Parkosidis et al. (2025) proposes a unified mass transfer process (GeMT) and eccentric mass transfer in binary systems to solve some of the shortcomings in CEE. The paper demonstrates that eccentric RLOF can act as a replacement for a CE phase, which allows angular momentum to be removed from the system and decrease the binary's orbital period as material flows out of the outer RL, L_2, and leaves system entirely.

Parkosidis et al. (2025) also demonstrates that GeMT in binary systems can stop the potential merger of the binary objects in CEE, as a slightly eccentric binary could survive the contraction and then potentially merge later in its lifetime.

6 CONCLUSIONS

WD binaries can evolve through a phase of SRLOF, Common Envelope, or a hybrid of both. Current literature as discussed in this review discusses the stability of the mass transfer process via RLOF and/or a CE phase and debate on which constraints should be placed on different types of WD binary systems to explain and model those evolutionary phases correctly.

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